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ABSTRACT

Keywords:
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Conclusion: Pediatric thyroplasty requires technical adaptations. Use of a cartilage implant allows observation of medialization stability during the growth. These findings are particularly relevant of contraindication or failure of nonselective reinnervation.

Patients with less than one year of follow-up were excluded. Data collected included demographics, paralysis etiology, nature of symptoms, duration of progression before thyroplasty, details of the

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perioperative period, and postoperative and long-term functional outcome evaluations.

2.2. Surgical technique

The surgical technique derived from the Isshiki type 1 thyroplasty technique with some adaptations to pediatric anatomy [9].

The procedure was performed under general anesthesia, preceded by endoscopy to assess passive mobility of the paralyzed vocal cord and eliminate cricoarytenoid ankylosis. Intraoperative ventilation was ensured by a laryngeal mask allowing endoscopic control of the medialization during the procedure.

The cervical incision was made in a neck fold overlying the thyroid ala. The infrahyoid muscles were retracted laterally to expose the thyroid ala. The location of the height of the glottic plane and the dimensions of the cartilage window were determined by placing 27-gauge needles through the thyroid cartilage (Fig. 1). The cartilage window was delimited between the anterior limit of the vocal cord and the middle part of the vocal process of the arytenoid. In the case of cricoarytenoid ankylosis, the posterior part of the window was delimited by the posterior macula flava to allow medialization of the cord.

After sectioning the cartilage window, the internal perichondrium of the thyroid ala was detached using otological instruments to mobilize it. The window was then depressed into the paraglottic space under endoscopic control to determine the proper position of the graft to obtain a paramedian positioning of the vocal fold (Fig. 2). If additional cartilage was necessary to obtain satisfactory medialization, a fragment of cartilage was removed from the upper edge of the ipsilateral thyroid ala, cut to size, and fixed perpendicularly between the inner thyroid ala and the graft (Fig. 3). The surgical steps are illustrated in Fig. 4. The cartilage grafts were fixed using PDS 3.0 (Ethicon Inc., New Jersey/USA) and the incision was closed without drainage. Feeding was resumed on the procedure day and discharge was permitted if respiratory status and nutrition were satisfactory.

2.3. Morphological evaluation

Laryngeal fiberoptic laryngoscopy was performed at postoperative months 1, 6, and 12, and then annually to check the vocal cord position and glottic competence during phonation. Morphological evaluation was completed with laryngeal ultrasound, allowing visualization of the cartilage graft position and vocal cord.

2.4. Functional evaluation

Functional evaluation was performed preoperatively and at postoperative months 1, 6, and 12, and annually thereafter. The functional assessment was by parental evaluation using a visual analogue scale

(VAS) for aspiration and respiratory congestion and dyspnea. Voice assessment was based on the Grade, Roughness, Breathiness, Asthenia, and Strain (GRBAS) score measured at each visit. The number of bronchopulmonary infections was counted before and after the intervention.

2.5. Statistical analysis

Pre- and postoperative assessments were compared using Student's *t*-test for paired samples, with $p < 0.05$ considered statistically significant. SPSS v.14 (IBM, USA) was used for statistical calculations and graphical representation of the results.

3. Results

3.1. Demographic data

The study included 11 patients aged <10 years, seven boys and four girls, with a median age of 26 months (range: 8–115 months) at the time of surgery. All patients had unilateral laryngeal paralysis, the etiology of which was iatrogenic in all cases, related to cardiovascular surgery in six cases, brain stem tumor resection in three cases, esophageal atresia in one case, and mediastinal lymphoma resection in one case. The three patients with brain stem tumors also had involvement in X and IX, as well as involvement in XII in two cases and VII in one case. Median duration of paralysis progression before surgical management was 17 months (range: 6–103 months). Patient demographics are summarized in Table 1.

All patients had severe aspiration, chronic bronchopulmonary congestion, and recurrent bronchopulmonary infections with an average of 2.6 pneumonia episodes (range: 1–4) prior to surgical management. Dyspnea was reported in eight patients, all of whom were considered moderate.

Preoperative laryngeal electromyography was performed in seven patients, all of whom showed signs of severe denervation associated with a loss of rhythmicity relative to ventilation.

3.2. Postoperative results

The proposed technique achieved cord medialization in all patients. Medialization was obtained through the cartilage window alone in four patients, all of whom were aged <5 years. In the other cases, the addition of a single complementary fragment of thyroid ala cartilage was necessary. The three patients with high vagal damage benefited from an additional hypopharyngoplasty procedure during the same operation.

No intra or postoperative complications were observed. Feeding was resumed at postoperative day 0, except in patients who had undergone hypopharyngoplasty, among whom it was deferred until postoperative day 3. The mean time of hospitalization was 2.4 days (range: 2–5), with

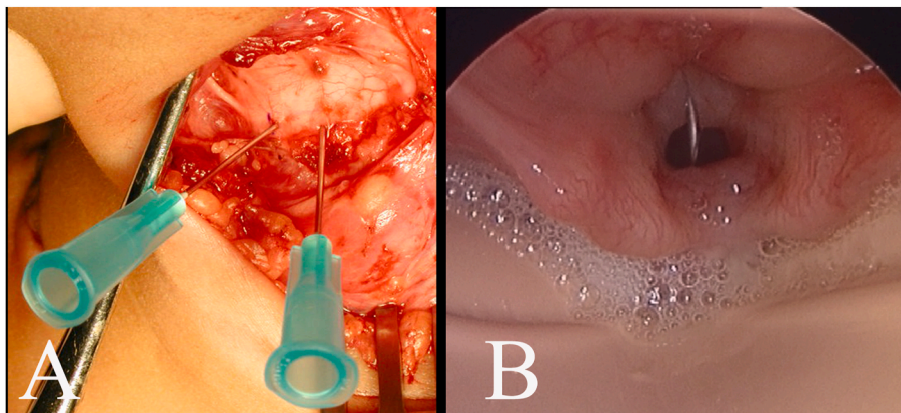


Fig. 1. Locating the glottic plane. Left thyroplasty. A 27-gauge needle is passed through the thyroid cartilage to identify the height and posterior limit of the glottic plane, allowing definition of the cartilage window dimensions (A). The location is facilitated by use of the laryngeal mask, which allows endoscopic control of the needle positioning, here in the anterior commissure to define the anterior limit of the window (B).

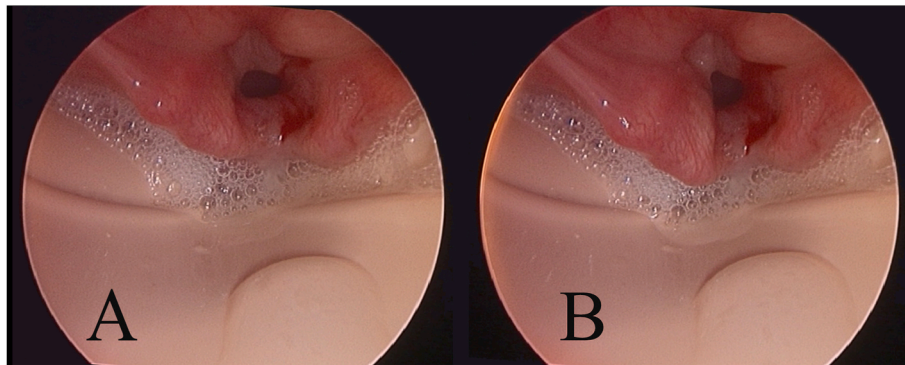


Fig. 2. Endoscopic control of medialization

Left thyroplasty. Use of the laryngeal mask allows the medialization to be controlled from the initial position (A) to the final position after adaptation of the cartilage implant (B).

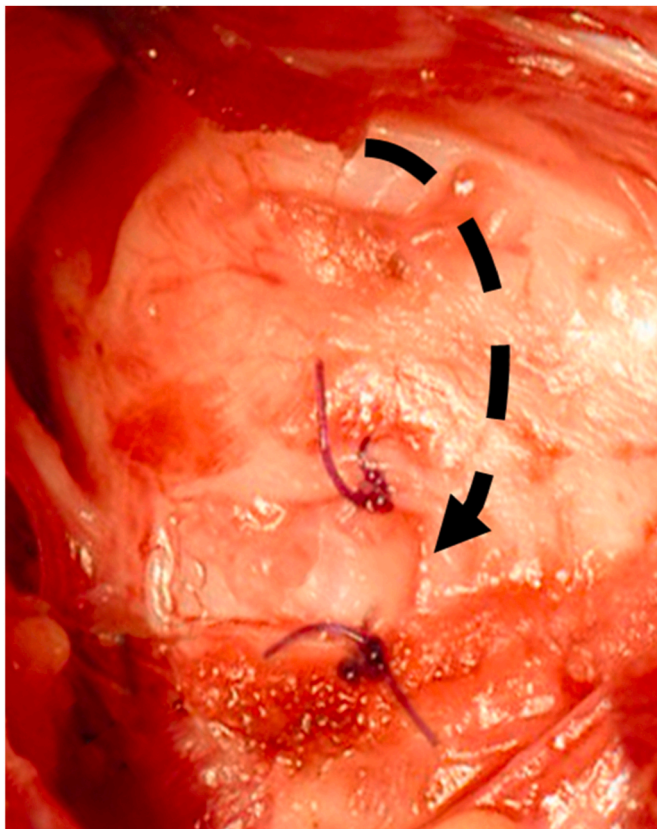


Fig. 3. Implant adaptation

Once the cartilage window is placed, if it is necessary to increase the medialization, a cartilage rod is taken from the upper edge of the ipsilateral thyroid ala and refined on demand according to endoscopic data. The cartilage is highly flexible at this age, making it necessary to fix this rod perpendicularly to avoid its mobilization.

a longer monitoring duration in patients who had undergone hypopharyngoplasty.

Immediate postoperative evaluation identified a nearly complete disappearance of aspiration, with a median change in VAS from 9/10 to 0/10 ($p < 0.0001$). Concerning bronchopulmonary congestion, the median VAS changed from 8/10 to 0/10 ($p < 0.0001$). No recurrence of bronchopulmonary infection was observed. There was no increase in dyspnea, with the median VAS remaining 1/10. Voice evaluation showed a significant improvement, with the median GRBAS score decreasing from 8 (range: 6–11) to 2 ($p < 0.0001$). The functional results

are summarized in Fig. 5.

3.3. Outcomes

Long-term evolution, to a median follow-up of 77 months (range: 17–122), showed stable results on all parameters in 10 cases. One patient presented a late phonatory degradation requiring a vocal fold injection of hyaluronic acid 9 years after thyroplasty. Ultrasound follow-up showed no resorption of the cartilage implant and no deformation of the thyroid alar with time. Absence of standardization of cartilage graft measurements with ultrasound did not allow to assess objectively the cartilage growth.

4. Discussion

While cord medialization by thyroplasty is commonly used in adults, few studies have reported its use in children, with only eight studies involving a total of 34 patients, the majority of whom were adolescents [1–8]. Table 2 provides a summary of these data. In the present study, we report the only experience with the use of cartilage implant in children.

Among the technical characteristics of thyroplasty, three issues appear specific to young children: the location of the height of the glottic plane, the nature of the implant, and the methods of intraoperative control of the medialization.

Identification of the height of the glottic plane and the definition of the height and dimensions of the window in the thyroid ala are critical to the success of the technique. The position of the glottic plane in children is not standardized. Link et al. showed that implant placement based on markers used in adults was too high in relation to the ventricular band in seven out of eight patients [3]. The location of the glottic plane is therefore an essential issue in pediatric thyroplasty. Computed tomography evaluation is impossible at this age, as the absence of cartilage calcification disallows identification of the limits of the thyroid cartilage. Laryngeal ultrasonography can be valuable because it allows ready identification of the thyroid cartilage, paraglottic space, and glottic plane in the axial plane, making it possible to define the length of the window and thickness of the implant necessary for sufficient medialization. However, using the coronal plane section, identification of the height of the glottic plane in relation to the lower edge of the thyroid ala remains difficult. Identifying the height of the glottic plane and the definition of the limits of the cartilage window using a needle passed through the cartilage, as described by Link et al. appears to be the most reliable and reproducible technique [3]. Nevertheless, this technique risks cord injury and paraglottic hematoma, which may be responsible for postoperative dyspnea.

Regarding the choice of implants, a variety of materials have been described for cord medialization. Silicone implants, led by the

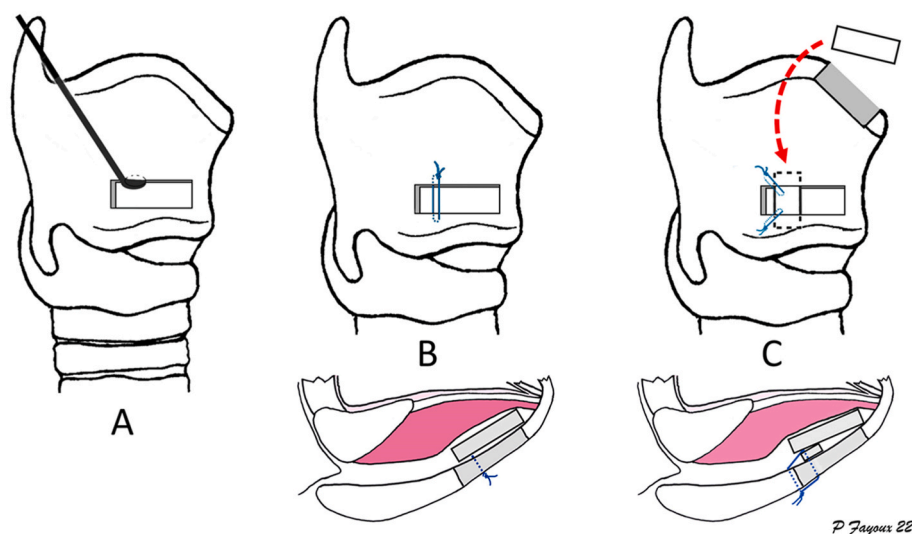
**Fig. 4.** Surgical procedure

Fig. 4 illustrates the surgical procedure. After sectioning the cartilage window, the internal perichondrium of the thyroid wing ala was detached using otological instruments (A). The window was then impacted into the paraglottic space under endoscopic control. The cartilage graft is maintain with a single suture between the edges of the cartilage window (B). If additional cartilage was necessary to obtain the proper medialization, a fragment of the upper edge of the thyroid ala is harvested, seized and fixed perpendicularly between the inner thyroid ala and graft (C).

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Table 1

Demographics. Summary of each patient's clinical data and postoperative course. The three patients with X-stem paralysis related to a neurosurgical injury (brain stem tumor) underwent simultaneous hypopharyngoplasty.

N	Sex	Age (months)	Etiology	Side	Duration evolution (months)	Associated procedure	Discharge delay (days)	Follow-up duration (months)	Revision
1	M	20	CVS	L	19	0	2	36	0
2	M	18	NC	L	6	1	3	30	0
3	F	8	CVS	L	7	0	2	103	0
4	M	115	CVS	L	103	0	2	77	0
5	F	11	NC	R	7	1	5	42	0
6	M	26	NC	L	6	1	3	17	0
7	M	13	CVS	L	10	0	2	122	Injection
8	M	38	AO	L	37	0	2	105	0
9	F	61	CVS	L	56	0	2	52	0
10	M	73	Chest lymphoma	R	21	0	2	87	0
11	F	46	CVS	L	17	0	2	82	0

M: Male, F: Female, CVS: Cardiovascular surgery, NC: Neurosurgery, AO: Esophageal atresia, R:right, L: Left, 0: No, 1: Yes.

Montgomery system of preformed implants, are usually used in adults [10]. However, this implant line does not have a model suitable for children. Instead, silicone implants are custom-made from Silastic blocks, based on intraoperative measurements [2–5,8]. Among other synthetic materials, Gore-Tex strips have been used; their flexibility and ability to modify position within the paraglottic space as guided by simultaneous flexible laryngoscopy make the cartilage window's position less critical [7,8]. However, the major disadvantage of synthetic implants is the lack of growth capacity, which requires repeated changes of implants when used in young children [3,8].

In contrast to synthetic implants, in their type 1 thyroplasty technique, Isshiki et al. proposed to use the cartilage window as an implant by placing it in the paraglottic space [9]. Medialization was completed, if necessary, by an additional cartilage rod. The theoretical advantage of using cartilage in children lies in its growth capacity. Although Sasaki et al. described a case of resorption of the cartilage window in a 64-year-old adult, our observations tend to show that in children, the cartilage implant does not resorb and shows relatively homogeneous growth [11]. The survival of cartilage grafts has also been demonstrated in laryngeal reconstruction [12–15]. Experimental studies demonstrated that the grafted cartilage survived well after implantation in the vast majority of cases. Pashley et al. demonstrated the survival and growth of the grafted cartilage in a child on a histological specimen [16]. The survival of thyroid cartilage graft in laryngotracheal reconstruction was also reported in clinical studies, showing stable middle term results, suggesting a growth capacity [17,18]. These studies also reported the

absence of thyroid cartilage deformation after graft removal.

Although pediatric thyroplasty under local anesthesia may be considered in some adolescents, ideally supplemented by infiltration of the superficial plexus, it is usually performed under general anesthesia. Control of the medialization is thus more difficult under these conditions. The use of endoscopic control through a laryngeal mask, also proposed by Gardner et al. seems the most suitable solution to optimize vocal cord placement. It allows control for achieving optimal medialization, while taking care to remain slightly paramedian position in young children, to avoid the risk of inducing dyspnea [4].

Complications in pediatric thyroplasty appear to be rare. Sipp et al. reported one case of postoperative pneumonia and Wolter et al. reported a case of transient stridor rapidly regressing on corticosteroids [7,8]. Analysis of the summary of published cases (Table 2) indicates that initial results are satisfactory overall and comparable with alternatives (i.e., medialization by intracordal injection and nonselective reinnervation) [7,19]. Butskiy et al. also reported best results on aspiration with thyroplasty compared with injections in children aged <10 years (19).

On the other hand, the duration of improvement with the use of synthetic implants remains limited, with a degradation reported to occur between 6 months and 2 years, including in adolescents. This degradation systematically requires an iterative implant change or additional injections [2,4,20]. A major limitation of our study is the follow-up duration with only 6 patients with a duration more than 5 years, than didn't allow to assess long term evolution of the graft, particularly during the pubertal growth period. Nevertheless, our results showed

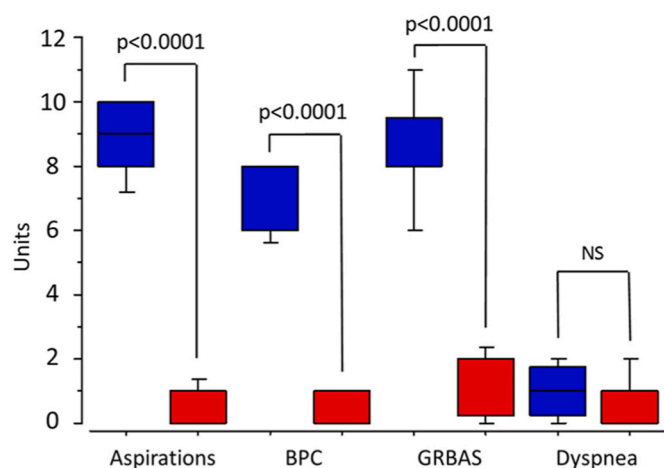


Fig. 5. Functional outcomes

Summary of pre- (blue) and postoperative (red) evaluation of functional parameters. Aspiration, bronchopulmonary congestion (BPC), and dyspnea were evaluated by parental visual analogue scale evaluation from 0 (no symptoms) to 10 (permanent or severe symptomatology). Evaluation of the voice patch was based on the GRBAS score. Significant improvements in these parameters were observed, without increased dyspnea. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 2

Summary of current publications addressing pediatric thyroplasty.

First author (publication year)	Number of patients	Age (years)	Implants	Results
Tucker (1986) (1)	1	NS	NS	NS
Isaacson (1990) (2)	1	14	Silastic	Voice degradation and aspirations at 6 months
Link (1999) (3)	8	2, 2, 7, 12, 14, 14, 14, 17	Silastic	Voice improvement in 5 out of 11 procedures Improvement of aspirations in 7 out of 9 procedures
Gardner (2000) (4)	2	4, 8	Silastic	Voice improvement at 4 and 12 months
Daya (2000) (5)	1	3	Silastic	Minimal voice enhancement
Shah (2006) (6)	4	NS	NS	Postoperative voice improvement in all patients Unspecified duration of effectiveness
Sipp (2007) (7)	3	5, 13, 18	Gore-Tex	Disappearance of aspirations Unspecified duration of effectiveness
Wolter (2019) (8)	14	Average age (DS) Silastic:16.5 (5.9) Gore-Tex: 4.5 (2.7)	11 Silastic 3 Gore-Tex	Silastic: improvement Voice 91%: term 1.8 years Aspirations 80%: duration 2.2 years Gore-Tex: improvement 100% voice: term 1.9 years Aspirations 50%: duration 1.3 years

NS: Not specified.

that the use of cartilage implant reduces the risk of reintervention since no patient required an implant change and only one required a complementary injection at the end of growth. This late degradation may be related to cord atrophy, which is not prevented by thyroplasty. Under these conditions, in the absence of laryngeal synkinesis by aberrant reinnervation, which would maintain cord trophicity, it seems preferable to use nonselective reinnervation (NSR) as a first-line treatment, which shows good long-term results in children, both vocally and for aspiration [7,20–22]. Based on these considerations, thyroplasty may be proposed in case of extensive prior neck surgery in whom donor nerves are not readily identifiable to perform a NSR, spontaneous partial reinnervation with residual glottic insufficiency or failure of prior NSR. We suspect that thyroplasty might be more effective than intracordal injection in cases of ankylosis with glottic gap, as the cord is probably more easily tensioned with an implant that can force the entire paraglottic space. In this case, it is nevertheless imperative that the implant should not be placed on the vocal process of the arytenoid.

5. Conclusion

Although thyroplasty is rarely used in young children, these results are potentially valuable. Nevertheless, optimizing these outcomes requires technical adaptations compared with the procedures described for adults. Use of a cartilage implant would allow obtaining a stability of medialization in most cases, reducing the risk of iterative interventions during growth. Its indications in young children seem to be particularly appropriate in cases with a severe gap. Given the satisfactory evolution during many years of the observed efficacy, this technique can be proposed at an early stage in cases of contraindication or failure of nonselective reinnervation. Nevertheless, a larger cohort study is needed to validate this indication and its long-term results until adulthood.

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